

Analysis of Hydrogen spectra using the Tracker Video

Aim:- To analyse the video of Hydrogen spectra in the Tracker tool and to find the prominent wavelengths of the Balmer series and to calculate the Rydberg's constant.

Apparatus: Computer with Tracker tool installed and recorded video of the spectrum.

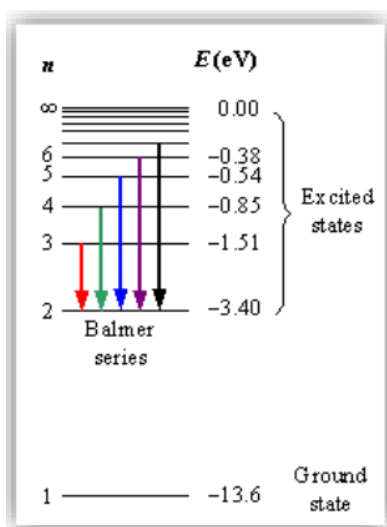
Principle:

The visible part of the hydrogen spectrum (Balmer series) originates from the transition of electron to $n_i = 2$ energy level from higher energy levels ($n_f = 3, 4, 5, \dots$) and the emitted wavelength is given by

$$\lambda = \frac{ch}{-E_1} \frac{(n_i^2 - n_f^2)}{n_i^2 n_f^2} = \frac{1}{R_\infty} \frac{(n_i^2 - n_f^2)}{n_i^2 n_f^2}$$

For Balmer Series $n_i = 2$ and $n_f = 3, 4, 5$

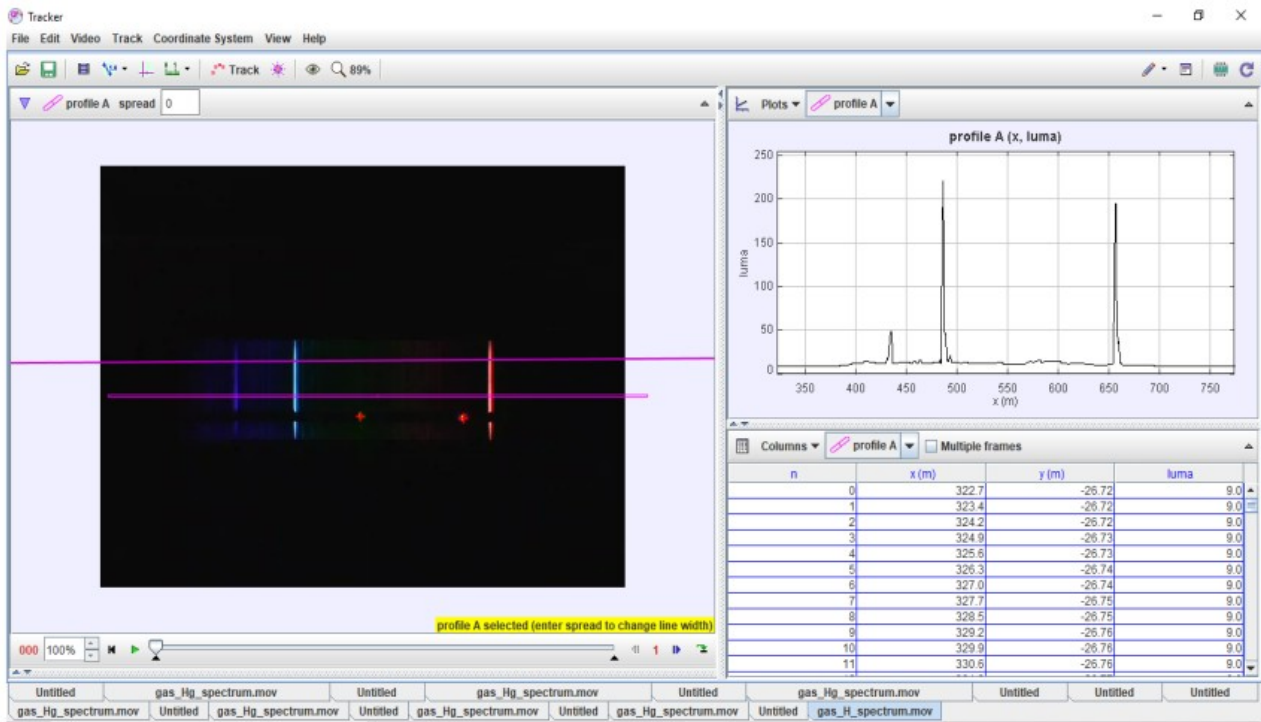
$$\text{Rydberg constant } R_\infty = \frac{1}{\lambda} \frac{(n_i^2 \cdot 4)}{n_i^2 - 4}$$



Procedure:

- Download and install tracker video in the computer from <https://physlets.org/tracker/>
- Import the recorded video of the spectrum(File > Open> File Chooser> select the file).
- To calibrate the spectrum, use the green and red laser pointers in the spectrum as the reference.
Track> New> Calibration Tools> Calibration Points> 'Shift+click' on the two laser pointers.
Give first calibration point x1-wavelength of green laser, 543nm
second calibration point x2-wavelength of red laser, 633nm
(Need not give the y values).
- To get the intensity profile, Track> New> Line Profile> 'Shift+click' on the left end of the spectrum and draw a horizontal line across the spectrum by dragging to the right end. The plot of wavelength vs intensity will be generated on the top-right column. Adjust the line on the spectrum by clicking and dragging across the slit to get the best profile.
- The peaks of the spectrum corresponding to the line can be obtained roughly by placing

the mouse pointer on the plot and accurately by analysing the tabulated values on the bottom-right column. Column 'x(m)' is the wavelength and 'luma' is the intensity.



Observations:

$n_f=2$

Line	Color	n_i	Wavelength λ (nm)	$R_\infty = \frac{1}{\lambda} \frac{(n_i^2 \cdot 4)}{n_i^2 - 4}$
H_α	Red	3		
H_β	Blue-Green	4		
H_γ	Blue	5		
H_δ	Violet	6		

Results:

The wavelengths of the prominent lines of the Balmer series are determined and hence calculated the Rydberg constant, $R_\infty = \dots \dots \dots \text{m}^{-1}$